Radian Measure and Arc Length

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Converting between degrees and radians

Convert each angle to radians, in exact form:

a 45° **b** 120° **c** 270° **d** 330°

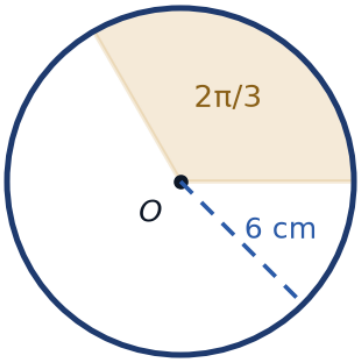
2 [Foundational] • Target: Converting between degrees and radians

Convert each angle to degrees. Round to one decimal place where needed:

a $\frac{\pi}{6}$ **b** $\frac{5\pi}{4}$ **c** $\frac{7\pi}{3}$ **d** 3.5 rad

3 [Foundational] • Target: Arc length and sector area

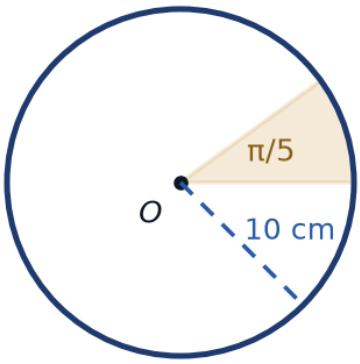
A circle has radius 6 cm. Find the exact length of the arc subtended by a central angle of $\frac{2\pi}{3}$.



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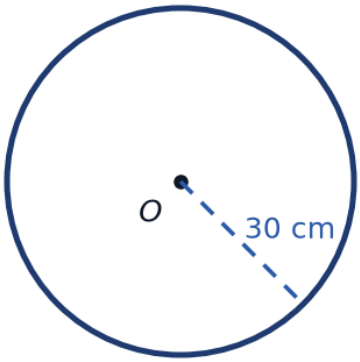
4 [Foundational] • Target: Arc length and sector area

A sector of a circle of radius 10 cm has central angle $\frac{\pi}{5}$. Find its exact area.



5 [Foundational] • Target: Arc length and sector area

A wheel of radius 30 cm rolls along the ground without slipping.



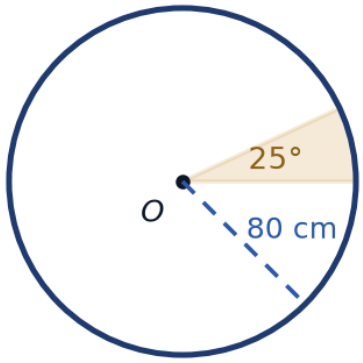
a Through what angle, in radians, does the wheel turn when it travels 1.5 m?

b Express this angle to the nearest tenth of a degree.

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6 [Foundational] • Target: Arc length and sector area

A pendulum of length 80 cm swings through an angle of $25°$. Find the length of the arc traced by its tip, to the nearest tenth of a centimetre.



7 [Foundational] • Target: Angular and linear speed

A carousel horse is 4 m from the center and completes one full rotation every 20 seconds.

a Find the angular speed of the horse, in radians per second.

b Find the linear speed of the horse, in m/s.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Angular and linear speed

A car's wheel has radius 35 cm and spins at 400 revolutions per minute. Find the car's speed in km/h, to one decimal place.

9 [Intermediate] • Target: Angular and linear speed

Two gears mesh together: the smaller gear has radius 5 cm and spins at 300 rpm. If the larger gear has radius 15 cm, find its rotational speed in rpm (the linear speed at the point of contact is the same for both gears).

10 [Intermediate] • Target: Coterminal angles and reference angles

Find a positive coterminal angle (between 0 and 2π) for each:

a $\frac{9\pi}{4}$ **b** $-\frac{\pi}{3}$ **c** $\frac{13\pi}{6}$

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11 [Intermediate] • Target: Coterminal angles and reference angles

Find the reference angle for each angle:

a $\frac{5\pi}{6}$ **b** $\frac{7\pi}{4}$ **c** $\frac{4\pi}{3}$

12 [Intermediate] • Target: The radian definition itself

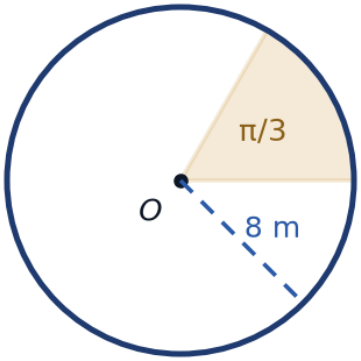
State the definition of one radian in terms of arc length and radius, and explain why radians are considered a 'natural' unit for angles.

13 [Intermediate] • Target: The radian definition itself

Explain why the full circle corresponds to 2π radians, using the circumference formula $C = 2\pi r$.

14 [Intermediate] • Target: Combined arc length and sector area problems

A sprinkler sprays water over a sector of radius 8 m and central angle $\frac{\pi}{3}$. Find both the area watered and the length of the outer arc of the watered region.

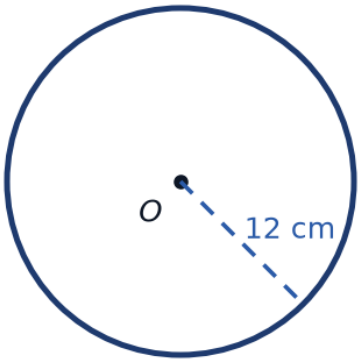


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Combined arc length and sector area problems • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A slice of pizza (a sector) has radius 12 cm and arc length 10 cm along its crust. Find the central angle in radians and the area of the slice.



16 [Challenging] • Target: Common angles in both units (reference table) • Cross-Lesson: 7.02 Logarithms and Their Properties

Complete a conversion table for the common angles $0°, 30°, 45°, 60°, 90°, 180°, 360°$ in radians.

17 [Challenging] • Target: Satellite and orbital speed applications • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its angular speed in rad/h and its linear speed in km/h, to the nearest whole number.

18 [Challenging] • Target: Satellite and orbital speed applications • Cross-Lesson: 7.04 Applications: Growth and Decay

Earth rotates once every 24 hours. Find the linear speed (due to rotation alone) of a point on the equator, using Earth's radius ≈ 6371 km, to the nearest km/h.

19 [Challenging] • Target: Negative angles and full-turn arcs • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the exact arc length swept by a radius-4 circle for a rotation of $-\frac{3\pi}{2}$ radians (clockwise), and interpret the sign.

20 [Challenging] • Target: Common angles in both units (reference table) • Cross-Lesson: 8.02 Graphing Rational Functions

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Degrees and Radians Conversion

Convert 150° to exact radians in simplest form.

a. $\frac{5\pi}{6}$

b. $\frac{2\pi}{3}$

c. $\frac{3\pi}{4}$

d. $\frac{7\pi}{6}$

22 [Foundational] • Target: Degrees and Radians Conversion

Convert $\frac{7\pi}{4}$ radians to degrees.

a. 315°

b. 225°

c. 300°

d. 330°

23 [Foundational] • Target: Degrees and Radians Conversion

Find a positive angle coterminal with $-\frac{2\pi}{3}$ in the interval $[0, 2\pi)$.

a. $\frac{4\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{5\pi}{3}$

d. $\frac{2\pi}{3}$

24 [Foundational] • Target: Arc Length

State the formula for arc length s subtended by a central angle θ (in radians) in a circle of radius r.

a. $s = r\theta$

b. $s = \frac{1}{2}r^2\theta$

c. $s = 2\pi r\theta$

d. $s = \frac{r}{\theta}$

25 [Foundational] • Target: Arc Length

Find the arc length intercepted by a central angle of $\frac{2\pi}{3}$ in a circle of radius 9 cm.

a. 6π cm

b. 3π cm

c. 18π cm

d. 12π cm

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26 [Foundational] • Target: Arc Length

An arc of length 10 m subtends a central angle of 2 radians. What is the radius of the circle?

a. 5 m

b. 20 m

c. 10 m

d. 2.5 m

27 [Foundational] • Target: Sector Area

State the formula for the area A of a sector of a circle with radius r and central angle θ (in radians).

a. $A = \frac{1}{2}r^2\theta$

b. $A = r^2\theta$

c. $A = \frac{1}{2}r\theta$

d. $A = \pi r^2\theta$

28 [Foundational] • Target: Sector Area

Find the area of a sector with radius 6 in and central angle $\frac{\pi}{4}$.

a. $\frac{9\pi}{2}$ sq in

b. 9π sq in

c. $\frac{9\pi}{4}$ sq in

d. 18π sq in

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Sector Area

If a sector of radius 4 cm has an area of 8π cm^2, find its central angle θ.

a. π radians

b. 2π radians

c. $\pi/2$ radians

d. 4π radians

30 [Intermediate] • Target: Linear and Angular Speed

What is the relationship between linear speed v, radius r, and angular speed ω (in rad/time)?

a. $v = r\omega$

b. $v = \frac{\omega}{r}$

c. $v = r^2\omega$

d. $v = 2\pi r\omega$

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31 [Intermediate] • Target: Linear and Angular Speed

A wheel of radius 2 ft rotates at 60 rpm. Find its angular speed in radians per second.

a. 2π rad/s

b. 4π rad/s

c. π rad/s

d. 120π rad/s

32 [Intermediate] • Target: Linear and Angular Speed

In the previous question ($r = 2$ ft, $\omega = 2\pi$ rad/s), what is the linear speed of a point on the rim?

a. 4π ft/s

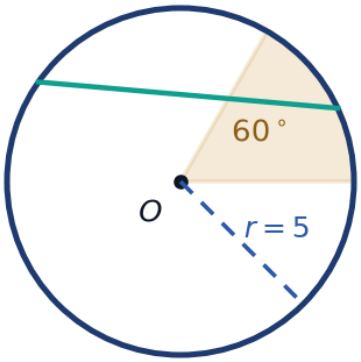
b. 2π ft/s

c. 8π ft/s

d. 120 ft/s

33 [Intermediate] • Target: Linear and Angular Speed

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?



a. 6 units

b. $6\sqrt{3}$ units

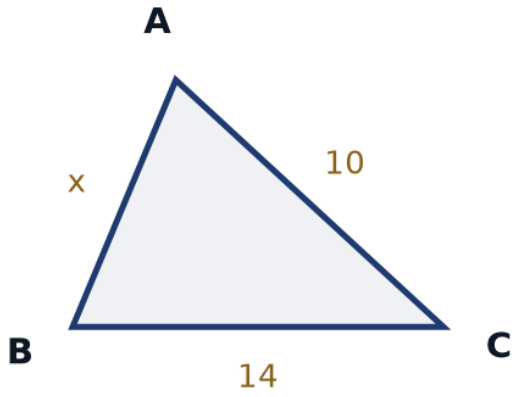
c. 12 units

d. $12\sqrt{3}$ units

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34 [Intermediate] • Target: Converting between degrees and radians

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?



a. 10.3

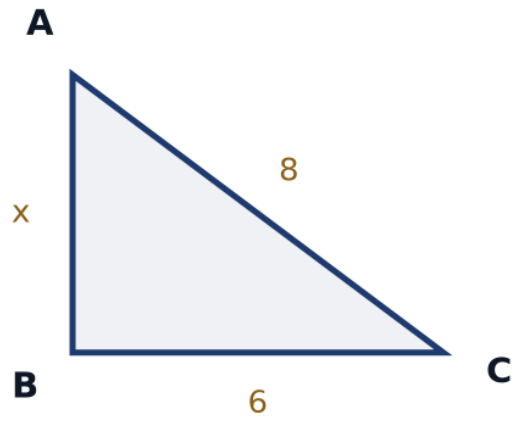
b. 12.1

c. 14.8

d. 8.5

35 [Intermediate] • Target: Arc length and sector area

In a right-angled triangle *ABC*, the angle *B* is 90 degrees. If the length of side *AB* is 8 units and the length of side *BC* is 6 units, what is the length of the hypotenuse *AC*?



a. 10 units

b. 14 units

c. 48 units

d. 20 units

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36 [Intermediate] • Target: Angular and linear speed

In triangle ABC, side $a = 10$ units and side $b = 15$ units. If the angle $C = 60^\circ$, find the length of side c. Round your answer to two decimal places.

a. 13.23

b. 14.70

c. 12.96

d. 15.52

37 [Intermediate] • Target: Coterminal angles and reference angles

In triangle ABC, the lengths of sides a, b, and c are opposite to angles A, B, and C respectively. Given that $a = 7$, $b = 10$, and angle $C = 60^\circ$, find the exact length of side c.

a. $\sqrt{79}$

b. 9

c. $\sqrt{91}$

d. 10

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The radian definition itself • Cross-Lesson: 7.04 Applications: Growth and Decay

Convert $\frac{2\pi}{3}$ radians to degrees.

a. 120°

b. 240°

c. 150°

d. 60°

39 [Challenging] • Target: Combined arc length and sector area problems • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Convert $\frac{5\pi}{4}$ radians to degrees.

a. 225°

b. 255°

c. 210°

d. 135°

40 [Challenging] • Target: Common angles in both units (reference table) • Cross-Lesson: 8.02 Graphing Rational Functions

Convert $\frac{7\pi}{6}$ radians to degrees.

a. 150°

b. 210°

c. 105°

d. 240°

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41 [Challenging] • Target: Satellite and orbital speed applications • Cross-Lesson: 8.03 Introduction to Limits

Convert $\frac{\pi}{3}$ radians to degrees.

a. 300°

b. 60°

c. 90°

d. 30°

42 [Challenging] • Target: Negative angles and full-turn arcs • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Convert $\frac{\pi}{4}$ radians to degrees.

a. 30°

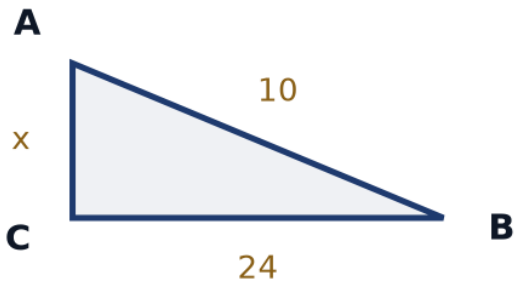
b. 75°

c. 45°

d. 315°

43 [Challenging] • Target: Degrees and Radians Conversion • Cross-Lesson: 9.02 Geometric Sequences and Series

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.



a. 25

b. 26

c. 27

d. 28

44 [Challenging] • Target: Arc Length • Cross-Lesson: 9.03 The Binomial Theorem

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

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45 [Challenging] • Target: Sector Area • Cross-Lesson: 1.01 Functions and Function Notation

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. 25

b. 50

c. 100

d. 125

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